DESIGN DOCUMENT

IFRS TAXONOMY & CURRENCY-AWARE NORMALIZATION

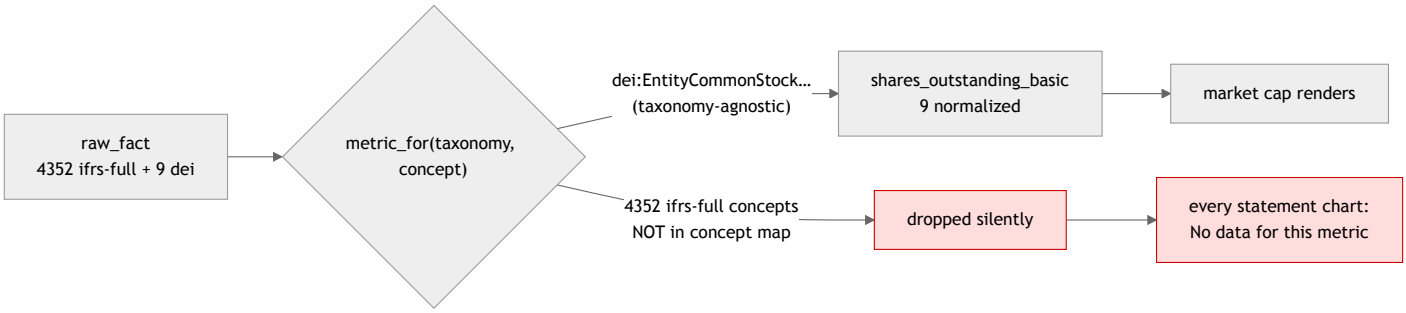
Status: Approved 2026-07-11 — recommended architecture adopted in full (incl. §5.2 debt policy and §6 currency scope). See §9. **Companions:** `docs/ifrs-currency-support-techspec.md` **Parent:** `docs/architecture.md` **Trigger:** INFY (Infosys Ltd, CIK 0001067491) renders a blank dashboard — 4361 raw facts ingested, only 9 normalized.

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1. Overview

Infosys files with the SEC as a **foreign private issuer**: its financial facts are tagged in the `ifrs-full` taxonomy, not `us-gaap`. Ingestion succeeds (748 filings, 4361 raw facts, 9 price points), but the normalization layer maps **only** `us-gaap` (51 curated concepts) plus a single `dei` concept. It contains **zero** `ifrs-full` mappings, so 4352 of 4353 IFRS facts are silently dropped. The one metric that survives — `shares_outstanding_basic` — comes through the taxonomy-agnostic `dei:EntityCommonStockSharesOutstanding` candidate, which is why market cap renders while every statement chart shows "No data for this metric."

Investigating the fix surfaced **three distinct gaps**. Only one is the blank screen; the other two are latent correctness risks that this document exists to separate, triage, and resolve durably.



The key structural finding: **the normalization engine is already taxonomy-agnostic.** `orchestrator.rs` calls `concept_map::metric_for(&f.taxonomy, &f.concept)` and normalizes whatever the table returns, with no `us-gaap` assumption anywhere on the path. IFRS produces nothing today for exactly one reason — the table has no IFRS rows. This is what makes Gap 1 additive rather than structural.

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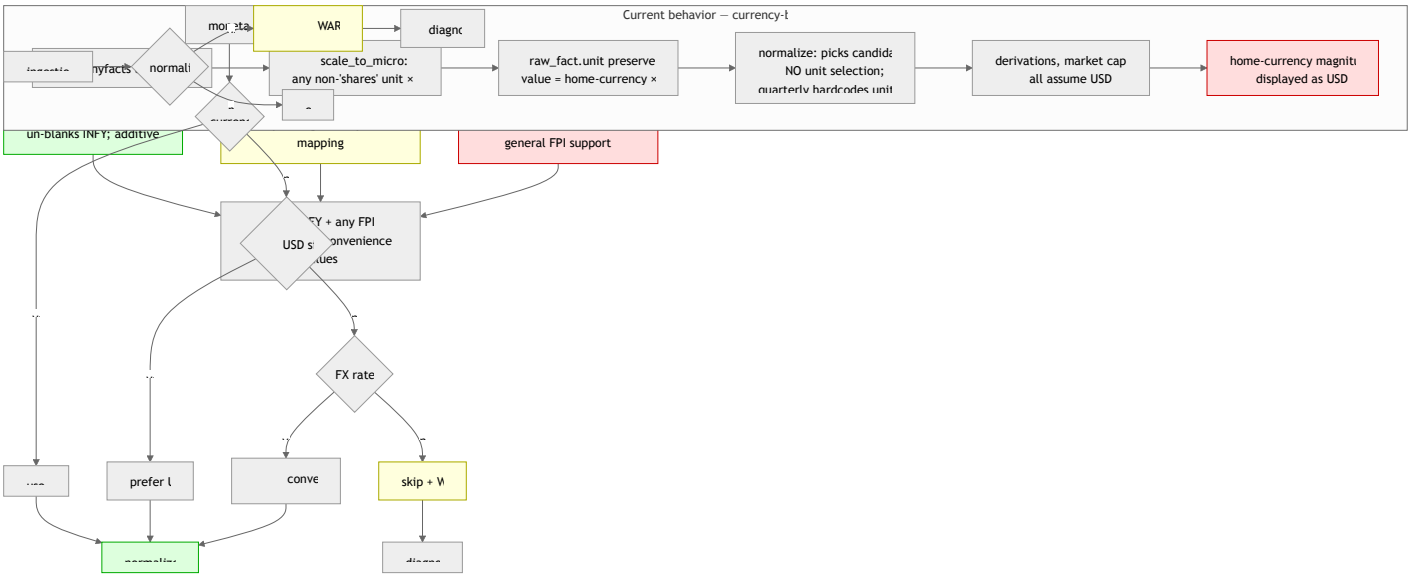
2. The three gaps

#	Gap	Root cause	Blocks INFY?	Correctness risk if ignored
1	**IFRS concept coverage** (incl. concept	No `ifrs-full` rows in `concept_map.rs`	**Yes — this is the blank screen**	—

#	Gap	Root cause	Blocks INFY?	Correctness risk if ignored
	drift across IFRS versions)			
2	**Debt semantics under IFRS 16**	IFRS capitalizes all leases as `LeaseLiabilities`; issuer carries no traditional borrowings	No	Mapping leases → debt inflates leverage and breaks cross-taxonomy comparability
3	**Currency-blind pipeline**	`scale_to_micro` and the normalize path treat every monetary unit as USD	No — INFY files USD convenience translations	**High** — any FPI without USD convenience values yields silently wrong numbers labelled USD

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3. Triage — what must be fixed, and when



- **Gap 1 — must fix now.** This is the reported bug. The fix is additive (add `ifrs-full` candidates to the existing per-metric arms) and unblocks INFY plus every IFRS filer that provides USD convenience translations.
- **Gap 2 — decide now, implement cheaply.** Not a code-architecture problem; a product/domain policy. It does not block the dashboard (INFY is debt-free). But the policy **must be decided before any IFRS debt concept is mapped**, or we bake in a silent cross-taxonomy inconsistency.
- **Gap 3 — guard now, full fix before general FPI support.** INFY is saved only by luck: Infosys tags USD convenience values in its 20-F, which the pipeline scales correctly ($3.162B `ProfitLoss` FY25, $10.94B `Revenue` FY18 both verified accurate). The moment an issuer reports only in its home currency, today's pipeline stores home-currency magnitudes labelled USD — silently wrong. We add a **loud diagnostic now** (so this never manifests as a blank/garbage screen again) and schedule the FX-conversion fix as its own work package.

Do not fix: nothing here is a no-op. All three warrant action; they differ in urgency and altitude.

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4. Gap 1 — IFRS concept coverage

4.1 What the data actually supports

I matched every internal metric against the `ifrs-full` concepts Infosys actually reports (counts are fact rows in the production DB). **14 of ~16 statement metrics have a clean, directly-mappable IFRS concept:**

Internal metric	IFRS concept (fact count)	Note
Revenue	`RevenueFromContractsWithCustomers` (21), `Revenue` (3)	**concept drift** — modern IFRS 15 tag vs. legacy; needs both candidates
CostOfRevenue	`CostOfSales` (24)	clean
GrossProfit	`GrossProfit` (24)	clean
OperatingIncome	`ProfitLossFromOperatingActivities` (24)	clean — issuer *does* tag it
NetIncome	`ProfitLoss` (24)	clean
EpsBasic / EpsDiluted	`Basic/DilutedEarningsLossPerShare` (24)	clean, USD/shares
SharesOutstandingBasic	`WeightedAverageShares` (24)	clean
SharesOutstandingDiluted	`AdjustedWeightedAverageShares` (24)	naming: "adjusted" = diluted
CashAndEquivalents	`CashAndCashEquivalents` (28)	clean
TotalAssets / TotalLiabilities	`Assets` (16) / `Liabilities` (16)	clean
TotalEquity	`EquityAttributableToOwnersOfParent` (16), `Equity` (32)	parent-first, incl-NCI fallback (mirrors US-GAAP ordering)
CashFromOperations	`CashFlowsFromUsedInOperatingActivities` (24)	clean
CapitalExpenditures	`PurchaseOfPropertyPlantAndEquipmentClassifiedAsInvestingActivities` (9)	sign already normalized by `apply_sign`
DepreciationAmortization	`DepreciationAndAmortisationExpense` (24)	clean
PropertyPlantAndEquipmentNet	`PropertyPlantAndEquipment` (32)	clean

The read-time derivations need **no change**: free cash flow (`ProfitLoss` + D&A – capex), the capex PP&E roll-forward fallback (`PropertyPlantAndEquipment` + D&A), and operating margin all resolve from the metrics above once the

concepts are mapped.

4.2 Options

Option A — inline `ifrs-full` candidates in the existing arms. *(Recommended.)* Extend each `concepts_for` arm with its IFRS candidates after the `us-gaap` ones. Keeps a single source of truth per metric; the ordered-candidate mechanism already handles multi-taxonomy fallback; `metric_for` reverse lookup already iterates all candidates. ~20 concept lines.

Option B — externalize the map to a data file (TOML/CSV) loaded at startup. More scalable for many taxonomies, but introduces a load path, a parse-error surface, and drift between file and `Metric` enum. Not justified at two taxonomies.

Concept drift within a taxonomy (e.g. IFRS 15 changed `Revenue` → `RevenueFromContractsWithCustomers`) is handled by the **same ordered-candidate list** we already use for US-GAAP synonyms — no new mechanism.

4.3 Recommendation

Option A, plus two **forward-looking guards** (§7) so a future metric can't silently regress IFRS coverage and so a coverage collapse never again surfaces as a blank screen.

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5. Gap 2 — debt semantics under IFRS 16

5.1 The problem

Under **IFRS 16** every lease is capitalized as a `LeaseLiabilities` line. Infosys carries essentially no financial borrowings (`Borrowings`: 1 fact) — its balance-sheet "debt" is almost entirely lease liabilities (22 facts). Under **US-GAAP (ASC 842)** operating-lease liabilities are also on-balance-sheet, but our `us-gaap` map deliberately routes them to **neither** `LongTermDebt` **nor** `CurrentDebt` — only interest-bearing borrowings count as debt.

If we map IFRS `LeaseLiabilities` → debt, an IFRS filer's leverage would include operating leases that an otherwise-identical US filer's would exclude. That is a **cross-taxonomy comparability defect**: the same metric would mean two different economic things depending on where the company files.

5.2 The governing principle & options

> **Economic-equivalence rule:** a metric means the same economic thing regardless of taxonomy. An IFRS concept may be mapped to a metric only if it is economically equivalent to that metric's US-GAAP concept — not merely "close."

Option A — map interest-bearing borrowings only; exclude leases. *(Recommended.)* `LongTermDebt` ← `NoncurrentBorrowings` / `Borrowings`; `CurrentDebt` ← `CurrentBorrowings`. Leases are excluded, exactly as for US-GAAP. Preserves comparability. For Infosys this yields ~zero debt — which is *correct*.

Option B — count leases as debt on both sides. Consistent across taxonomies, but changes existing US-GAAP numbers (larger blast radius), and conflicts with how most screeners report "debt." Rejected as default.

Option C — introduce a distinct `LeaseLiabilities` metric. Cleanest long-term: surface leases explicitly rather than forcing them into "debt." Deferred — adds a metric and UI surface; not required to render INFY correctly.

5.3 Recommendation

Adopt the economic-equivalence rule as written policy. Implement **Option A** now. Keep **Option C** on the roadmap for when lease disclosure becomes a product requirement.

Approved (2026-07-11). Lease liabilities are **not** debt: only interest-bearing borrowings map to `LongTermDebt` / `CurrentDebt`, matching the US-GAAP treatment. For Infosys this correctly yields ~zero debt. A separate `LeaseLiabilities` metric (Option C) is deferred until lease disclosure is a product requirement.

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6. Gap 3 — currency-aware normalization (the architectural core)

6.1 Why today's pipeline is currency-blind

Three concrete facts from the code:

- `scale_to_micro` (`companyfacts.rs:118`) multiplies **any** unit except `shares` by 10^6 — it never inspects the currency.
- The normalize path (`orchestrator.rs`) selects candidates and best-per-period **without ever filtering on unit**; the quarterly branch hardcodes `unit: "USD"` (line 185); the annual/instant branches preserve `f.unit` but everything downstream assumes USD.
- `normalized_fact` already has `fx_rate_micro`, `fx_rate_source`, `fx_rate_date`, `original_value`, `original_unit` columns and there is an `fx_rate` table — **the FX scaffolding exists but is wired to `None` everywhere.**

INFY escapes this only because Infosys files USD convenience translations (4032 of its facts are USD). A pure-INR filer would flow home-currency magnitudes straight through as "USD."

The `require_usd` guard on the **market-data** adapter (`market_data.rs:106`) already embodies the correct principle for quotes — *reject rather than mislabel*. Statement facts have no equivalent guard. The fix extends that principle to the fact pipeline.

6.2 Options

Option A — USD-preference candidate selection. When a concept is tagged in multiple currencies for a period, prefer the USD-unit fact and drop non-USD when a USD sibling exists. Cheap; fully fixes convenience-translation filers (INFY) and prevents accidental home-currency pickup. **Does not** help pure-foreign-currency filers.

Option B — FX conversion at normalization. *(The durable fix.)* For a monetary fact whose currency ≠ USD, convert to USD using an FX rate as-of the period end, store the converted value, and record provenance in the **existing** columns (`original_value` , `original_unit` , `fx_rate_micro` , `fx_rate_source` , `fx_rate_date`). Requires an `FxRateSource` and populating the `fx_rate` table.

Option C — reject + diagnostic. If a monetary fact is non-USD and cannot be converted, **skip it and emit a WARN ingestion event** rather than store a mislabelled number. This is the accuracy-preserving floor and aligns with the project's hard-accuracy rule.

6.3 Recommended architecture — a layered currency-resolution stage

Introduce an explicit **currency-resolution step** between raw-fact selection and normalized-fact persistence, applying a strict ladder per monetary fact:

Supporting structural changes:

1. **Treat currency as first-class.** `raw_fact.unit` already carries it; stop assuming USD downstream. The quarterly branch must stop hardcoding `unit: "USD"` and instead flow the resolved currency.
2. **Split `scale_to_micro` 's two responsibilities** — "is this monetary?" (scale ×10^6) vs. "which currency is it?" (drives resolution). Scaling stays; currency detection becomes explicit.
3. **`FxRateSource` trait** feeding the `fx_rate` table, mirroring the existing `MarketDataSource` adapter shape. Rate keyed by (currency, date); resolution uses the period-end date.
4. **Provenance is mandatory** whenever a value is FX-converted: the converted USD value is the fact; `original_value` / `original_unit` / `fx_rate_*` record how we got there, upholding the product's auditability principle.

6.4 Recommendation

Ship **Option A + Option C now** (they are cheap, and together they make the pipeline *honest* — correct for convenience-translation filers, loud-and-safe for everyone else). Ship **Option B** as the follow-on work package that unlocks general FPI support. Sequencing and the FX-source choice are in the tech spec.

Approved (2026-07-11). The layered resolution ladder (prefer USD → FX-convert with recorded provenance → skip + diagnostic) is the target architecture. WP3a (Options A + C) lands with WP1; **WP3b (Option B, FX conversion) is scheduled as its own follow-on unit of work, not this cycle.**

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7. Forward-looking guards — stop silent recurrence

The deeper defect behind all three gaps is **silence**: 4352 dropped facts produced a blank screen with no error. These guards convert each failure mode into a loud, testable signal.

Guard	Prevents	Mechanism
Coverage-matrix test	Gap 1 regressions	Unit test over `Metric::ALL`: any metric with `us-gaap` candidates must also declare `ifrs-full` candidates (or be explicitly annotated US-only, e.g. bank concepts). Fails CI when a new metric forgets IFRS.
Raw-to-normalized ratio diagnostic	Gaps 1 & 3 in the wild	At end of ingestion, if `normalized_facts / raw_facts` falls below a threshold (e.g. the 9/4361 seen here), emit a **WARN** ingestion event naming the dominant unmapped taxonomy. The blank screen becomes a visible diagnostic.
Economic-equivalence review checklist	Gap 2	Documented rule (§5.2) + a checklist item on any concept-map change: "is this IFRS concept economically equal to the US-GAAP concept for this metric?"
Currency provenance assertion	Gap 3	Test: no `normalized_fact` may carry a non-USD `unit`; any FX-converted fact must have non-null `original_unit` + `fx_rate_*`.

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8. Decision summary

Gap	Verdict	Chosen approach	Status
1 — IFRS coverage	**Fix now**	Inline `ifrs-full` candidates (Opt A) + coverage-matrix test	**Approved**
2 — Debt semantics	**Decide now, implement cheaply**	Economic-equivalence rule; borrowings-only, exclude leases (Opt A); `LeaseLiabilities` metric deferred (Opt C)	**Approved — leases are not debt**
3 — Currency	**Guard now, full fix scheduled**	USD-preference + reject/diagnostic now (Opts A+C); FX conversion via `FxRateSource` next (Opt B)	**Approved — WP3a this cycle, WP3b follow-on**
Cross-cutting	**Fix now**	Four guards in §7	**Approved**

The blank screen closes with Gap 1 alone. Gaps 2 and 3 are the difference between "INFY works" and "IFRS filers work correctly and safely in general": Gap 1 is fixed immediately, Gap 2's policy is locked (leases are not debt) before any debt concept is mapped, and Gap 3's guards land now with its FX conversion scheduled as the next unit of work.

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9. Approvals

Date	Decision	Approved by
2026-07-11	Recommended architecture adopted **in full** — no deviations.	Product owner
2026-07-11	**Gap 2 (debt):** lease liabilities are **not** debt; map interest-bearing borrowings only, for parity with the US-GAAP treatment. `LeaseLiabilities` metric deferred.	Product owner
2026-07-11	**Gap 3 (currency):** layered resolution ladder is the target architecture. WP3a (USD-preference + reject/diagnostic) ships with WP1; **WP3b (FX conversion) is a scheduled follow-on, not this cycle.**	Product owner
2026-07-11	**Guards (§7):** all four adopted and ship with WP1.	Product owner

Implementation status: not started. Deferred at the product owner's request while a separate agent is editing source files, to avoid write conflicts. This document and its companion tech spec are the authoritative record until implementation begins.